Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Component

Meaning in my words How would you explain component to a friend?	**How can you tell?** If east and north are positive, which component is negative for a southeast trip?
Example or sketch What two arrows could you use to build a displacement of 3 m east and 4 m north?	**Non-example and why** A classmate counts the diagonal and both parts as three separate trips. What is counted twice?

Resultant

Meaning in my words How would you explain resultant to a friend?	**How can you tell?** Where does the single arrow start and end on a tip-to-tail drawing?
Example or sketch Walk 3 m east, then 4 m north. What single arrow joins your start and finish?	**Non-example and why** Someone labels the resultant 7 m. Which length did they add up?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Magnitude

Meaning in my words How would you explain magnitude to a friend?	**How can you tell?** For a displacement of 5 m west, what part gives the magnitude?
Example or sketch A vector has perpendicular parts of 3 m and 4 m. How long is the diagonal?	**Non-example and why** A partner gives the magnitude as −5 m. What is the minus sign doing?

Quadrant

Meaning in my words How would you explain quadrant to a friend?	**How can you tell?** Which way does an arrow point if its horizontal part is negative and its vertical part positive?
Example or sketch Sketch a vector with components +2 m and −3 m. Where is its tip relative to its tail?	**Non-example and why** Do two vectors with the same length always point into the same quadrant?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a right-angle journey; label x, y, and the resultant with its direction.



My marks mean

On assignment: Resultant

SECTION E / BACK / FINAL MEETING ONLY Name _____

A tracker sees a 3 km displacement toward Earth and 4 km sideways. The second telescope needs the whole movement.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Navigation engineer**

On a map, give the camera operator one resultant arrow with a size and direction. Label the two component legs.

☐ **Map presenter**

Deliver a map update. Use one arrow for the displacement and show why “7 km closer to Earth” is the wrong caption.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor’s check: State the angle reference before explaining your direction.